

LAPTOP PRICE

1. OBJECTIVE

- **Predict Prices:** Estimate the probable price of a laptop based on its technical specifications. This helps companies determine appropriate pricing for new products, stay competitive in the market, and price according to customer expectations.
- **Identify Key Features:** Analyze which features have the most impact on pricing. This helps balance cost and performance in product design. Businesses can focus on the features that are most valuable to customers.
- **Market Segmentation:** Divide laptops into categories such as Budget, Premium based on customers needs and purchasing power. This enables companies to target the right customer segments and market their products more effectively, leading to increased sales.
- **Anomaly Detection:** Identify extremely cheap or expensive products that deviate from typical pricing trends. This plays a vital role in maintaining data quality, fixing listing errors, and preserving customer trust.
- **Support Strategy:** Use insights gained from data to make strategic decisions related to sales, inventory management, and product development. This helps businesses understand market demands, create better products, and gain a competitive advantage.

1.1 SCOPE

- **Identification of Key Factors Influencing Pricing:** This project analyzes which internal (such as technical specifications) and external (such as market trends) factors influence the price of a product. This information helps businesses decide which aspects they should focus on for better pricing decisions.
- **Market Segmentation:** Laptops are classified based on their features and prices to understand which products are suitable for which type of customers. This segmentation helps businesses reach their target customers more effectively.
- **Identification of Pricing Anomalies:** Some products are priced significantly higher or lower than the normal range, which may indicate data errors or listing mistakes. This project helps identify such cases so they can be corrected in time.
- **Product Classification by Price Category:** Laptops are categorized into price ranges such as Budget, Midrange, and Premium so that consumers can find suitable options based on their needs and budget, and businesses can position their products effectively in the market.

1.2 TECHNOLOGY

- Platform – Windows 11
Version – 23H2



- Tool – Excel
Version – 19H2



- Tool – Python
Version – 3.13



- Tool – Jupyter
Version – 7.3.2



2. DATA COLLECTION

2.1 DATA COLLECTION METHOD

1. Laptop Price

https://drive.google.com/file/d/1sU3TLwZA306IXScMvY_qgZs7-BvffSEw/view

2. Form of Data

- Data is in csv form having quantitative as well as qualitative data.

3. Rows & Columns

Rows: 1275 and columns: 25

2.2 DATA CLEANING & PRE-PROCESSING

- We used python library pandas. We import csv file into (df) dataframe. Using read_csv method.

```
import pandas as pd
df = pd.read_csv(r"C:\Users\Public\Downloads\laptop_prices.csv", encoding='utf-8-sig')
df.head()
```

	Company	Product	TypeName	Inches	Ram	OS	Weight	Price_euros	Screen	ScreenW	...	RetinaDisplay	CPU_company	CPU_freq	CPU_model
0	Apple	MacBook Pro	Ultrabook	13.3	8	macOS	1.37	1339.69	Standard	2560	...	Yes	Intel	2.3	Core i5
1	Apple	MacBook Air	Ultrabook	13.3	8	macOS	1.34	898.94	Standard	1440	...	No	Intel	1.8	Core i5
2	HP	250 G6	Notebook	15.6	8	No OS	1.86	575.00	Full HD	1920	...	No	Intel	2.5	Core i5 7200U
3	Apple	MacBook Pro	Ultrabook	15.4	16	macOS	1.83	2537.45	Standard	2880	...	Yes	Intel	2.7	Core i7
4	Apple	MacBook Pro	Ultrabook	13.3	8	macOS	1.37	1803.60	Standard	2560	...	Yes	Intel	3.1	Core i5

5 rows × 23 columns

- **Clean and Convert Ram from String to Integer**

```
df['Ram'] = df['Ram'].astype(str).str.replace('GB', '').astype(int)
```

Removes the 'GB' text from RAM values and converts the result to integer (e.g., '8GB' → 8).

- **Clean and Convert Weight from String to Float**

```
df['Weight'] = df['Weight'].astype(str).str.replace('kg', '').astype(float)
```

Strips 'kg' from strings like '2.2kg' → '2.2', and converts them into float values for analysis.

- **Convert Touchscreen to Binary (0/1)**

```
df['Touchscreen'] = df['Touchscreen'].apply(lambda x: 1 if x == 'Yes' else 0)
```

Changes 'Yes' to 1 and 'No' to 0.

- **Convert IPSpanel to Binary (0/1)**

```
df['IPSPanel'] = df['IPSPanel'].apply(lambda x: 1 if x == 'Yes' else 0)
```

Same idea as touchscreen: Encodes whether the screen uses IPS technology.

- **Encode RetinaDisplay to Binary (0/1)**

```
df['RetinaDisplay'] = df['RetinaDisplay'].apply(lambda x: 1 if x == 'Yes' else 0)
```

Binary format again, indicating presence of Apple's Retina Display.

- **Calculate PPI (Pixels Per Inch)**

```
df['PPI'] = ((df['ScreenW']**2 + df['ScreenH']**2)**0.5) / df['Inches']
```

Uses the Pythagorean theorem to calculate screen resolution (diagonal pixels), then divides by screen size in inches.

- **Create TotalStorage by Summing Primary and Secondary Storage**

```
df['TotalStorage'] = df['PrimaryStorage'] + df['SecondaryStorage']
```

Adds primary and secondary storage values to get total available storage per laptop.

- **Encode Screen Resolution into Numeric Levels**

```
df['Screen'] = df['Screen'].apply(lambda x: 1 if x == 'Full HD' else (2 if x == '4K Ultra HD' else 0))
```

Maps screen types to numbers:

0 → other/unknown

1 → Full HD

2 → 4K Ultra HD

- **One-Hot Encode Categorical Columns**

```
cat_cols = [col for col in ['Company', 'TypeName', 'OS', 'CPU_company', 'PrimaryStorageType', 'SecondaryStorageType', 'GPU_company'] if col in df.columns]
df = pd.get_dummies(df, columns=cat_cols, drop_first=True)
```

Converts categorical variables into binary (dummy) variables for machine learning models.

df.dtypes	
Product	object
Inches	float64
Ram	int64
Weight	float64
Price_euros	float64
Screen	int64
ScreenW	int64
ScreenH	int64
Touchscreen	int64
IPSPanel	int64
RetinaDisplay	int64
CPU_freq	float64
CPU_model	object
PrimaryStorage	int64
SecondaryStorage	int64
GPU_model	object
PPI	float64
TotalStorage	int64
Company_Apple	bool
Company_Asus	bool
Company_Chui	bool
Company_Dell	bool
Company_Fujitsu	bool
Company_Google	bool
Company_HP	bool
Company_Huawei	bool
Company_LG	bool

Company_Toshiba	bool
Company_Vero	bool
Company_Xiaomi	bool
TypeName_Gaming	bool
TypeName_Netbook	bool
TypeName_Notebook	bool
TypeName_Ultrabook	bool
TypeName_Workstation	bool
OS_Chrome OS	bool
OS_Linux	bool
OS_Mac OS X	bool
OS_No OS	bool
OS_Windows 10	bool
OS_Windows 10 S	bool
OS_Windows 7	bool
OS_macOS	bool
CPU_company_Intel	bool
CPU_company_Samsung	bool
PrimaryStorageType_HDD	bool
PrimaryStorageType_Hybrid	bool
PrimaryStorageType_SSD	bool
SecondaryStorageType_Hybrid	bool
SecondaryStorageType_No	bool
SecondaryStorageType_SSD	bool
GPU_company_ARM	bool
GPU_company_Intel	bool
GPU_company_Nvidia	bool
dtype: object	

Here, We checked data is null or not using isnill() function.

df.isnull().sum()	
Company	0
Product	0
TypeName	0
Inches	0
Ram	0
OS	0
Weight	0
Price_euros	0
Screen	0
ScreenW	0
ScreenH	0
Touchscreen	0
IPSPanel	0
RetinaDisplay	0
CPU_company	0
CPU_freq	0
CPU_model	0
PrimaryStorage	0
SecondaryStorage	0
PrimaryStorageType	0
SecondaryStorageType	0
GPU_company	0
GPU_model	0
dtype: int64	

3. METHODOLOGY

Data Collection:

- We have collected data from Google Drive.

Data Pre-processing & Cleaning:

- Converted the Ram column to integer by removing 'GB' from the values.
- Cleaned the Weight column by removing 'kg' and converting it to float.
- Calculated PPI (Pixels Per Inch) based on screen width, height, and inches.
- Combined PrimaryStorage and SecondaryStorage to create TotalStorage.
- Converted categorical columns like Touchscreen, IPSpanel and RetinaDisplay into binary (1/0) format.
- Mapped Screen to numerical values based on screen type (1 for Full HD, 2 for 4K Ultra HD).
- Applied one-hot encoding to categorical columns like Company, TypeName, OS, CPU_company, etc.

Used Algorithm:

- **Actual vs Predicted Laptop Prices:** Random Forest Regressor algorithm was used. It revealed that laptop prices depend on several factors. This model helps businesses understand which features have the greatest impact on pricing, enabling accurate price prediction.
- **Top 10 Important Features:** These features were derived from the feature importance scores of the Random Forest Regressor model. Identifying the main factors influencing pricing is crucial for businesses, as it helps them set appropriate product prices and meet customer expectations.
- **RAM vs Price:** KMeans Clustering algorithm was applied. It segmented laptops into two main categories, revealing a clear relationship between RAM and price. This is useful for businesses to identify which features and price points align with specific customer segments.

- **Weight vs Price:** Isolation Forest algorithm was used for anomaly detection. It identified laptops with unusually high or low prices that deviated from typical ranges. This is essential for businesses to detect and correct pricing errors or incorrect listings in a timely manner.
- **Laptop Price Category Classification:** Decision Tree Classifier was used to categorize laptops into price ranges (Budget, Midrange, Premium). It showed balanced performance across all categories. This helps businesses understand which features fit into specific price brackets & place products into appropriate categories based on customer needs.

4. FEATURE ENGINEERING

4.1 EXPLANATION OF FEATURE SELECTION AND EXTRACTION

- We analyzed various features such as Touchscreen, Retina Display, CPU frequency, RAM, Weight, Total Storage, PPI, Screen size, Company, CPU company, GPU company, Type Name, OS, Primary Storage Type, and Secondary Storage Type. By examining these features, we identified the most influential factors affecting laptop pricing. These insights enabled us to build models that understand the relative value of technical specifications in determining price.
- The importance of these features was determined from the model's feature importance scores, enabling businesses to make informed decisions on product pricing and customer satisfaction.
- Using features like RAM, Weight, and Price, we performed segmentation to group laptops into clusters. This revealed customer-preferred configurations and clear patterns in feature-price relationships, enabling more targeted marketing & pricing strategies based on performance tiers.
- With features like Weight, Screen size, Price, and Total Storage, we detected anomalies in pricing through outlier detection. This helped identify incorrect listings or mispriced products, improving data quality and customer trust.
- By categorizing laptops using key specifications such as RAM, Weight, CPU frequency, Primary Storage, and Secondary Storage, we accurately assigned products into Budget, Midrange, and Premium segments. These classifications supported better assortment planning and feature alignment with customer expectations.

4.2 DERIVED FEATURE

- We used StandardScaler to scale features like RAM, Weight, and Price because KMeans is a distance-based algorithm. If features have different scales, some features could disproportionately influence the clustering, leading to biased results. Scaling ensures that each feature contribute equally, leading to more balanced and accurate clustering of laptops into distinct categories.
- In Isolation Forest, we used StandardScaler to scale features like Weight, Inches, and Total Storage. Since the model detects anomalies based on the distance between points, scaling ensures that no feature (like storage or weight) dominates the outlier detection process. This makes the detection of true anomalies more reliable and ensures that the model is equally sensitive to all features.

- Although Decision Trees are not directly affected by feature scaling, we used StandardScaler to scale the features like RAM, Weight, and Storage for consistency. This ensures that all features are on the same scale, which can improve the interpretability of the tree and avoid potential biases when comparing features. Additionally, it helps in faster convergence and more robust model performance.

4.3 FEATURE SCALING

- PPI (Pixels Per Inch): To calculate screen resolution, which is derived from the screen width and height, and divided by Inches.

Formula: $df['PPI'] = ((df['ScreenW']**2 + df['ScreenH']**2)**0.5) / df['Inches']$

- TotalStorage: The sum of PrimaryStorage and SecondaryStorage, which represents the total storage of the laptop.

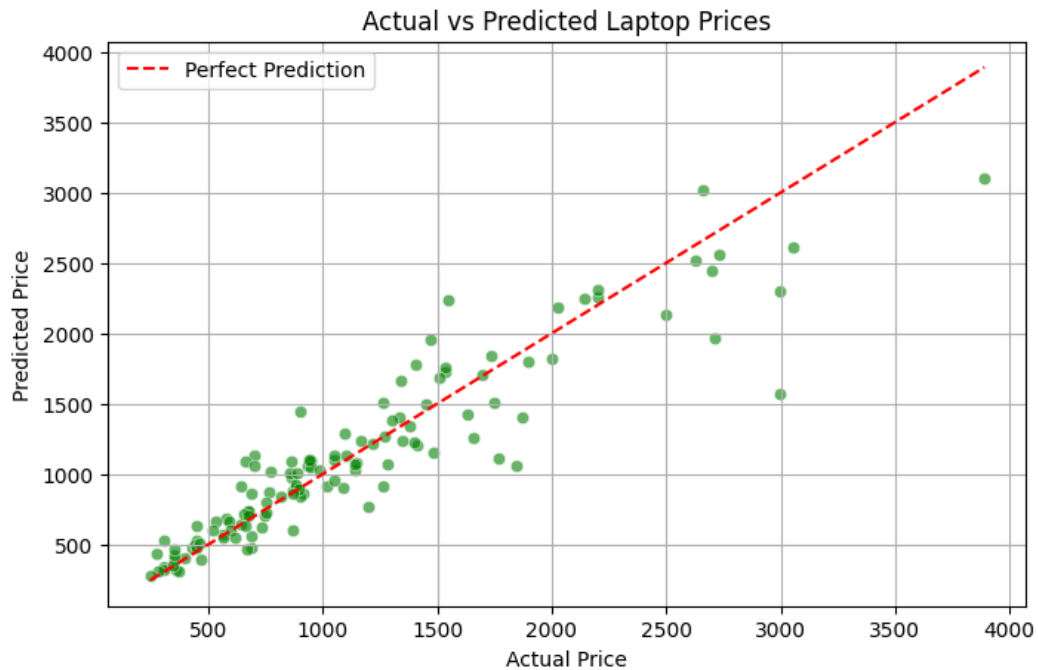
Formula: $df['TotalStorage'] = df['PrimaryStorage'] + df['SecondaryStorage']$

- Anomaly Detection: TotalStorage helps identify outliers with unusually high or low storage, marking them as potential anomalies.
- Price Prediction: PPI improves price prediction by reflecting screen quality.

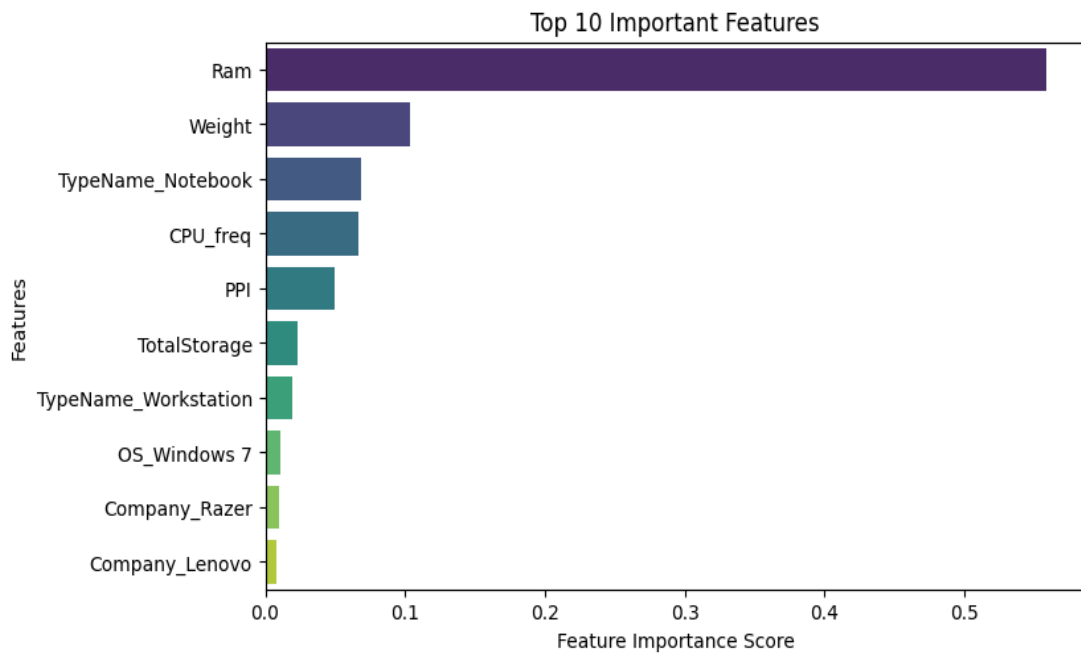
TotalStorage improves accuracy by representing storage capacity, both crucial for pricing.

5. Model Development

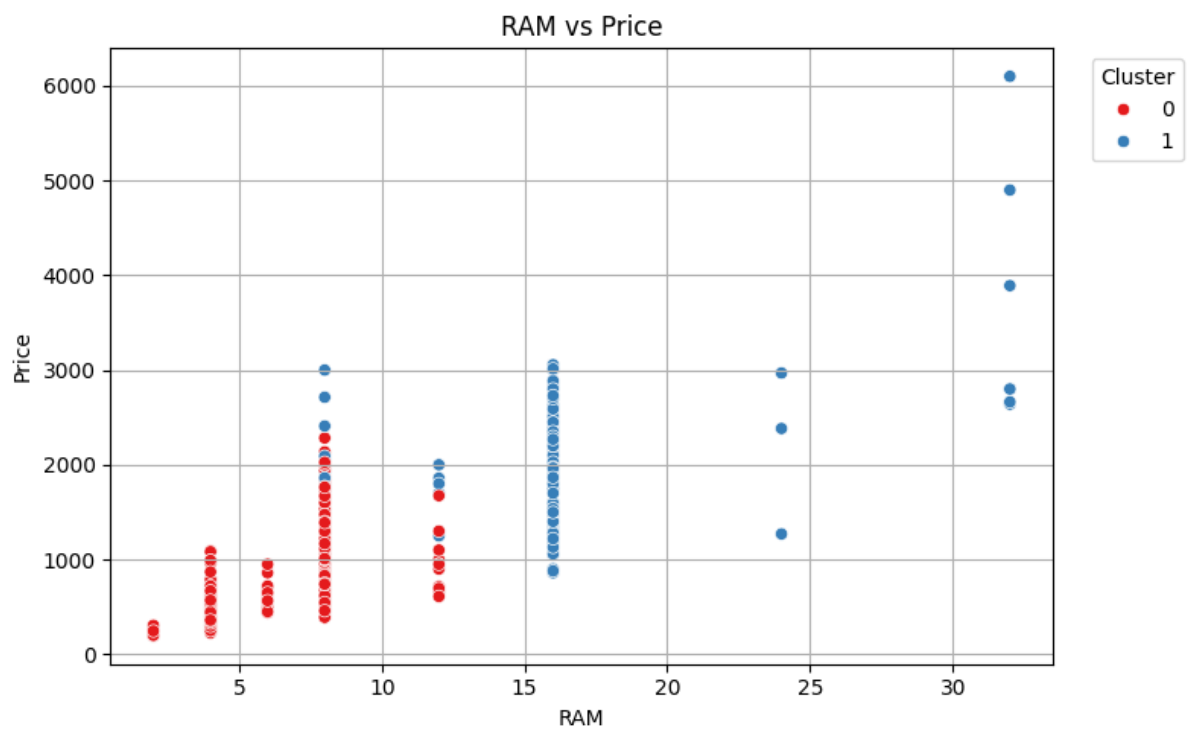
- Through an in-depth analysis of the laptop dataset, we uncovered key pricing drivers, product segmentation insights, and pricing inconsistencies. This exploration revealed how technical features influence pricing, allowing businesses to set competitive and customer-aligned prices. The insights also helped classify laptops into suitable price categories, identify unusual pricing behaviors, and understand customer-focused feature value. By leveraging these data-driven findings, companies can improve pricing strategies, optimize product offerings, and make informed decisions to drive sales and customer satisfaction.
- **Actual vs Predicted Laptop Prices:** The Random Forest model reveals that laptop prices depend on several factors. This model provides businesses with information on which features are key in influencing laptop prices, enabling accurate pricing.



- **Top 10 Important Features:** Identifying the key factors that influence laptop prices is crucial for businesses, as it helps them determine the prices of their products and meet customer needs.



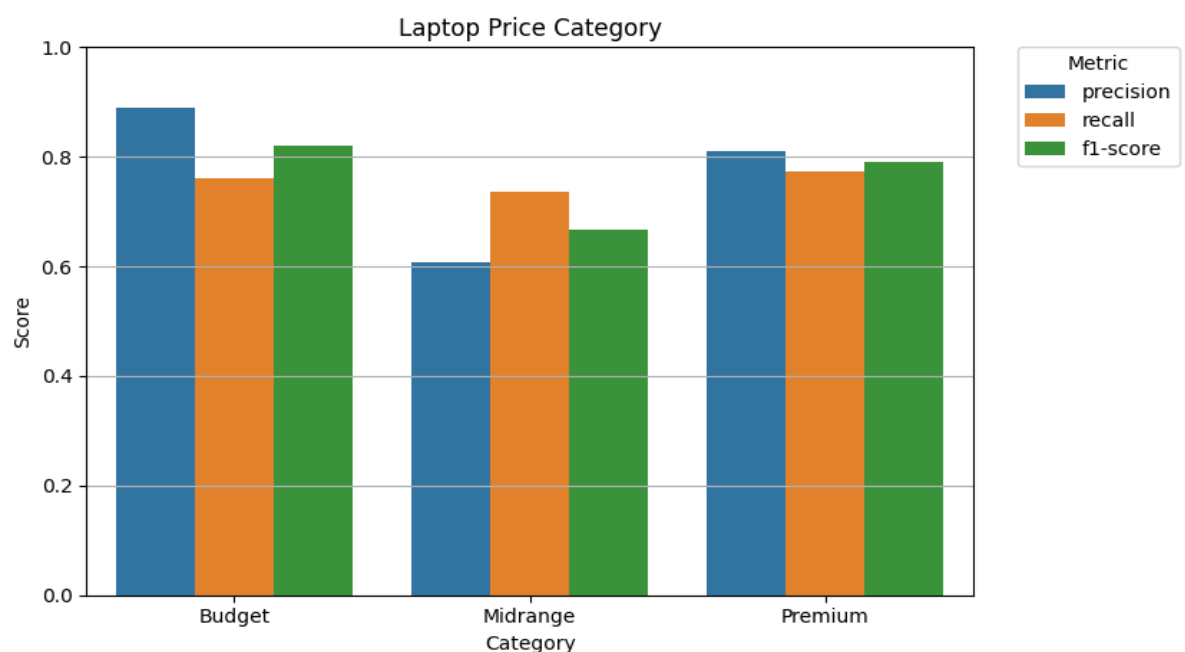
- **RAM vs Price:** KMeans clustering segmented laptops into two main categories, showing a clear relationship between RAM and price. This is useful for businesses as it helps companies understand which features and price points are most suitable for different customer segments.



- **Weight vs Price:** The Isolation Forest model identified laptops with abnormal prices, significantly higher or lower than normal. This is crucial for businesses as it helps identify products with potential pricing errors or incorrect listings, allowing for timely corrections.



- **Laptop Price Category Classification:** The Decision Tree model classified laptops into price categories, with balanced performance for each category. This helps businesses understand which features fit specific price categories and enables them to place products in the right category based on customer needs.



6. Evaluation

- The Random Forest model has an R^2 Score of 0.85, which indicates that the model has accurately captured 85% of the data's variability, meaning the predictions are quite good. The MSE is 75116.81, which shows a moderate error, but overall, the model's performance is quite good.

```
Random Forest:
MSE: 75116.81184432376
R2 Score: 0.8485578757096282
```

- The Silhouette Score of 0.5189 indicates moderate clustering quality. It suggests that the clusters have a decent separation, A score closer to 1 would indicate better-defined clusters.

```
Silhouette Score for KMeans with k=2: 0.5189167040458633
```

- The Decision Tree Classifier has an accuracy of 75.81%, which is good for the Budget and Premium classes, but the Midrange class has slightly lower precision (0.61). Overall, the model's performance is good.

```
Decision Tree Classifier:
Accuracy: 0.7580645161290323

Classification Report:
              precision    recall  f1-score   support

   Budget       0.89       0.76       0.82         42
  Midrange       0.61       0.74       0.67         38
   Premium       0.81       0.77       0.79         44

 accuracy              0.76         124
 macro avg       0.77       0.76       0.76         124
 weighted avg       0.77       0.76       0.76         124
```

7. CONCLUSION

7.1 SUMMARY OF ACHIEVEMENTS

- **Identification of pricing influencing factors:** Internal (such as RAM, processor, etc.) and external (such as market trends) elements were clearly found to affect laptop prices.
- **Support in data-driven decisions:** Insights obtained from the model contributed to improving strategies related to sales, inventory, and product development.
- **Identification of price anomalies:** Detection of extremely cheap or expensive products enhanced data accuracy and strengthened customer trust.
- **Product segmentation:** Clear classification into Budget, Midrange, and Premium categories based on RAM and other features made it easier to target the right customers.
- **Effective pricing strategy:** Presenting products at the right price not only gave companies a competitive edge in the market but also ensured customer satisfaction.